MAJOR PROJECT SYNOPSIS · 2025–26

Government Engineering College, Bidar

Computer Science & Engineering · VTU Belagavi

AI-Based Road Damage Detection

Municipal Maintenance Management System

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A photo alone does not resolve a complaint

Road damage needs a reliable path from observation to action.

Incomplete location

A photo or message may not identify the correct road segment.

Unclear priority

Separate reports make inspection and repair planning difficult.

Missing follow-up

Without a shared record, review, assignment and closure are hard to trace.

The project connects evidence, location, responsibility and repair status.

Four objectives guide the prototype

01	**Detect and locate**	Identify supported defects in images or sampled video frames; attach GPS and capture time.
02	**Prioritize inspections**	Estimate visible severity and explain why a case needs attention.
03	**Coordinate repairs**	Support officer review, team assignment and verified closure.
04	**Preserve road history**	Retain observations, decisions and repair evidence for future monitoring.

Initial scope: a defined road network, image uploads and batch video analysis.

Existing research supports the design

Research / resource	What it contributes	Design implication
Maeda et al. · 2018	Smartphone road imagery	Capture conditions matter.
Arya et al. · RDD2022	Annotated damage benchmark	Evaluate locally as well.
oracl4 · YOLOv8 project	Candidate model implementation	Check labels, licence and weights.

Our focus: connect detection to municipal review, repair and an auditable history.

How the system works

Capture & analyze

1 Upload photo/video + GPS

2 Detect potholes and cracks

3 Estimate type, severity and priority

Review & assign

4 Show photo and map on the dashboard

5 Officer verifies the result

6 Assign the approved case to a maintenance team

Repair & retain

7 Repair and upload after-repair photos

8 Officer checks and closes

9 Retain reports, repairs and road history

AI recommendations remain provisional until an officer reviews them.

The AI creates an evidence-backed report

Validate the input

Check file quality and capture time; sample usable video frames.

Run the detector

Retain the damage label, bounding box, confidence and model version.

Attach the location

Use capture GPS or a synchronized log; group repeated observations.

Target damage classes

Potholes
Longitudinal cracks
Transverse cracks
Alligator cracks

Subject to checkpoint support.

Missing GPS needs correction or a clearly labelled manual location.

Severity and priority are different

Severity

How concerning does the visible damage appear?

Propose low, medium or high concern from damage type and apparent extent.

Priority

Which case should be inspected first?

Combine severity, road importance, report age and reliable traffic exposure.

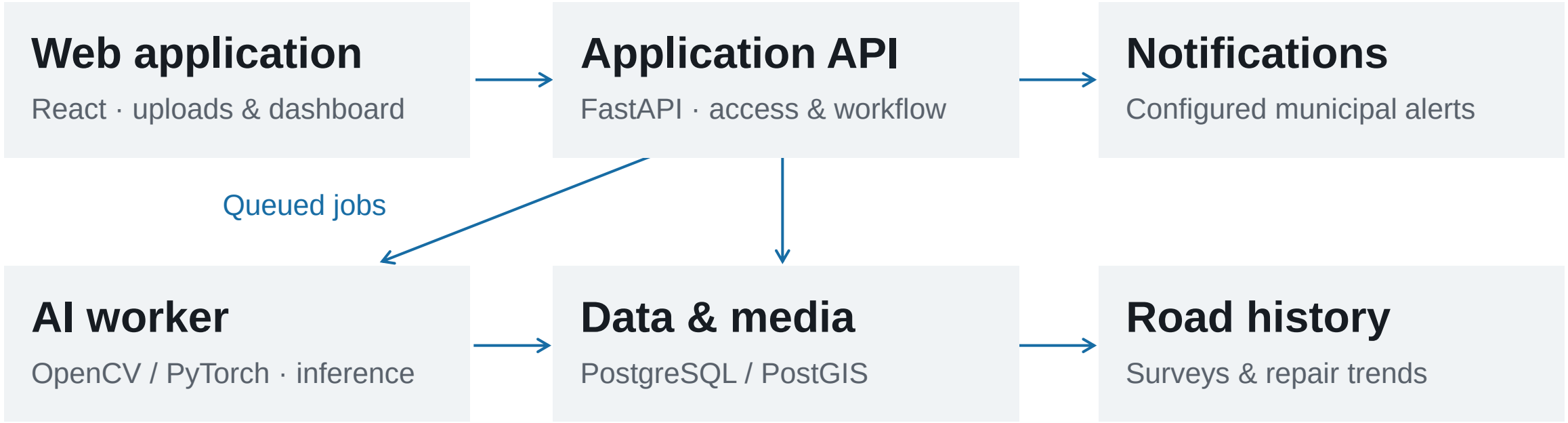
Road health

What was observed on this segment?

Use a provisional 0–100 score with survey date and coverage; higher means better.

Confidence is separate from severity. Officers can review scores and record overrides.

Proposed system architecture



Structured cases live in the database; original and annotated media live in object storage.

Officers control approval and closure

01	**Review the evidence**	Inspect the photo, map, predicted damage and suggested priority.
02	**Decide and assign**	Accept, correct, reject or merge; assign an approved case with a target date.
03	**Verify the repair**	Compare before/after evidence from the same location; close or return for more work.

Every decision retains the responsible user, timestamp and reason.

Build the prototype in four stages

1 Detection baseline

Confirm model classes; test annotated images and video sampling.

2 Incident pipeline

Connect uploads, GPS, duplicate grouping and provisional scores.

3 Municipal workflow

Add role-based review, assignment, repair evidence and history.

4 End-to-end validation

Measure quality and delay; test failures and closure controls.

Starting setup: GPS phone · 4-core PC · 16 GB RAM · ~50 GB SSD · optional GPU

Test the AI and the complete workflow

Evaluation area	Evidence to collect
Detection	Per-class precision, recall and mAP on held-out images
Location & duplicates	GPS/road-segment errors; missed or incorrect merges
Scoring & speed	Officer agreement; upload-to-result delay on stated hardware
Municipal workflow	Accept, reject, assign, repair, reopen and authorize closure

Keep repeated road views in one data split. Report actual results after testing.

A clear path to verified road repairs

One record connects each road report to a verified repair.

Expected outcome

Mapped incidents, clear ownership and before/after repair history.

Practical limits

Lighting, GPS quality and municipal participation affect reliability.

Future scope

Offline capture, regional languages and forecasting after sufficient verified history.

Next milestone: demonstrate one complete case from upload to officer-approved closure.